Market Basket Analysis & Customer Segmentation

Unlocking co-purchase patterns in children's products for actionable business insights.

Created by
Lashchinina Ekaterina and
Antropova Yana

Project Goal: Uncovering Co-Purchase Patterns

Our objective is to identify meaningful co-purchase patterns in children's products and and translate them into actionable business insights. We aim to answer key questions: questions:

Product Categories

Which product categories are purchased together?

Customer Segments

How do these patterns differ across across various customer segments? segments?

Cross-Sell & Bundles

Which associations are strong enough for cross-sell and bundled offers?

Data & Preparation: Transaction-Level Insights

We utilized transaction-level data from an online children's goods retailer. Each row represents one product within an order, ensuring granular detail.

Data Cleaning Process

- Delivered orders only
- No test/duplicate rows
- No partial returns

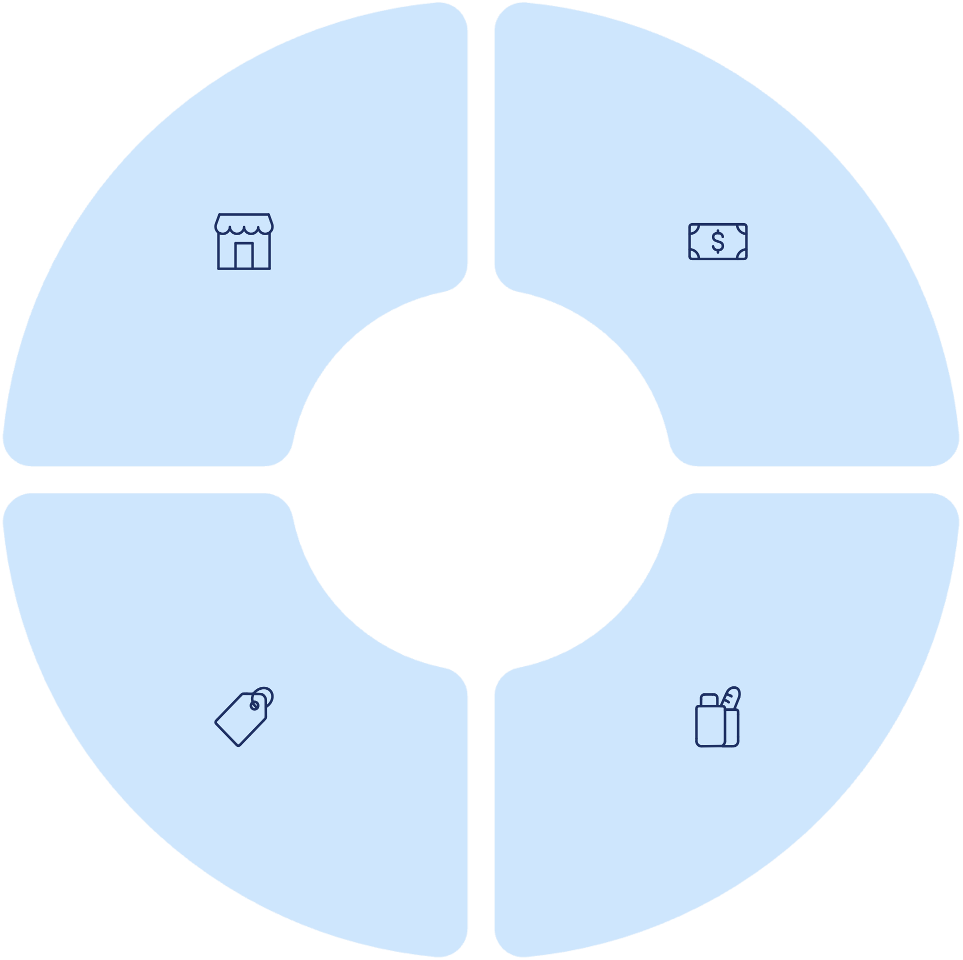
Categories were aggregated to the G2 level, including items like toys, diapers, toys, diapers, baby food, hygiene, and clothing, for a standardized analysis. analysis.

Customer Segmentation: K-Means Clustering

Customers were clustered into 9 distinct groups using the K-means algorithm. This segmentation is crucial because market basket analysis becomes much clearer becomes much clearer within homogeneous behavioral groups.

Purchase Frequency

Total Spend



Purchased Categories

Basket Size

Market Basket Analysis: Methods Used

To ensure robust and interpretable results, we applied three complementary methods within each customer cluster:

Apriori

Main method for interpretable rules (A → B).

FP-Growth

Faster alternative, confirming Apriori rule robustness.

ECLAT

Identifies frequent category pairs (A + B) via co-occurrence.

Using multiple methods significantly reduces the risk of identifying spurious patterns.

Key Association #1: Newborn Care Scenario

In Cluster 2, identified as "new parents," we observed a clear newborn-care shopping scenario.

Apriori / FP-Growth Rules

- **Baby care & hygiene → Baby food** (confidence ≈ 0.53, lift ≈ 1.53)
- **Baby food → Baby care & hygiene** (confidence ≈ 0.30, lift ≈ 1.53)
- **Diapers → Clothing & textile** (confidence > 0.80)

ECLAT Confirmed Pairs

- **Diapers + Clothing**
- **Diapers + Toys**

	antecedents	consequents	support	lift	confidence
0	(G2_КОСМЕТИКА/ГИГИЕНА)	(G2_ТОВАРЫ ДЛЯ КОРМЛЕНИЯ)	0.101835	1.529094	0.525994
1	(G2_ТОВАРЫ ДЛЯ КОРМЛЕНИЯ)	(G2_КОСМЕТИКА/ГИГИЕНА)	0.101835	1.529094	0.296041
2	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	(G2_ПОДГУЗНИКИ)	0.051510	1.334547	0.081384
3	(G2_ПОДГУЗНИКИ)	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	0.051510	1.334547	0.844660
4	(G2_КРУПНОГАБАРИТНЫЙ ТОВАР)	(G2_ИГРУШКИ)	0.061575	1.223879	0.153619
5	(G2_ИГРУШКИ)	(G2_КРУПНОГАБАРИТНЫЙ ТОВАР)	0.061575	1.223879	0.490566
6	(G2_ОБУВЬ)	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	0.248076	1.197129	0.757685
7	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	(G2_ОБУВЬ)	0.248076	1.197129	0.391955
8	(G2_ИГРУШКИ)	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	0.076377	0.961404	0.608491
9	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	(G2_ИГРУШКИ)	0.076377	0.961404	0.120674

Key Association #2: Toys as an Entry Point

Cluster 8 reveals that toys act as a significant entry point for customers, leading to purchases of everyday essentials.

Apriori / FP-Growth Rules

- **Diapers → Baby food** (lift ≈ 1.80)
- **Baby food → Diapers** (confidence ≈ 0.63)
- **Baby care & hygiene → Baby food**

ECLAT Confirmed Pairs

- Toys + Diapers
- Toys + Clothing

Interpretation: Toys attract customers, while everyday essentials expand the basket.

	antecedents	consequents	support	lift	confidence
0	(G2_ПОДГУЗНИКИ)	(G2_ДЕТСКОЕ ПИТАНИЕ)	0.037037	1.795213	0.106383
1	(G2_ДЕТСКОЕ ПИТАНИЕ)	(G2_ПОДГУЗНИКИ)	0.037037	1.795213	0.625000
2	(G2_ТОВАРЫ ДЛЯ КОРМЛЕНИЯ)	(G2_КОСМЕТИКА/ГИГИЕНА)	0.048148	1.300000	0.240741
3	(G2_КОСМЕТИКА/ГИГИЕНА)	(G2_ТОВАРЫ ДЛЯ КОРМЛЕНИЯ)	0.048148	1.300000	0.260000
4	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	(G2_ОБУВЬ)	0.059259	1.186813	0.175824
5	(G2_ОБУВЬ)	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	0.059259	1.186813	0.400000
6	(G2_ТОВАРЫ ДЛЯ КОРМЛЕНИЯ)	(G2_ПОДГУЗНИКИ)	0.081481	1.170213	0.407407
7	(G2_ПОДГУЗНИКИ)	(G2_ТОВАРЫ ДЛЯ КОРМЛЕНИЯ)	0.081481	1.170213	0.234043
8	(G2_КОСМЕТИКА/ГИГИЕНА)	(G2_ИГРУШКИ, G2_ТОВАРЫ ДЛЯ КОРМЛЕНИЯ)	0.033333	1.157143	0.180000
9	(G2_ИГРУШКИ, G2_ТОВАРЫ ДЛЯ КОРМЛЕНИЯ)	(G2_КОСМЕТИКА/ГИГИЕНА)	0.033333	1.157143	0.214286

Key Association #3: Stationery as a Basket Trigger

In Cluster 6, school-related items, specifically stationery and books, act as triggers for basket expansion.

Apriori / FP-Growth Rules

- **Stationery / books → Toys** (lift ≈ 1.57)
- **Stationery / books → Clothing & textile** (lift ≈ 1.54)

ECLAT Confirmed Pairs

- **Stationery + Toys**
- **Stationery + Clothing**

	antecedents	consequents	support	lift	confidence
0	(G2_КАНЦТОВАРЫ, КНИГИ, ДИСКИ)	(G2_ИГРУШКИ)	0.057613	1.573543	0.608696
1	(G2_ИГРУШКИ)	(G2_КАНЦТОВАРЫ, КНИГИ, ДИСКИ)	0.057613	1.573543	0.148936
2	(G2_КАНЦТОВАРЫ, КНИГИ, ДИСКИ)	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	0.032922	1.536759	0.347826
3	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	(G2_КАНЦТОВАРЫ, КНИГИ, ДИСКИ)	0.032922	1.536759	0.145455
4	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	(G2_ИГРУШКИ)	0.094650	1.081044	0.418182
5	(G2_ИГРУШКИ)	(G2_ТЕКСТИЛЬ, ТРИКОТАЖ)	0.094650	1.081044	0.244681
6	(G2_КОСМЕТИКА/ГИГИЕНА)	(G2_ПОДГУЗНИКИ)	0.275720	0.963031	0.626168
7	(G2_ПОДГУЗНИКИ)	(G2_КОСМЕТИКА/ГИГИЕНА)	0.275720	0.963031	0.424051
8	(G2_ТОВАРЫ ДЛЯ КОРМЛЕНИЯ)	(G2_ПОДГУЗНИКИ)	0.185185	0.833842	0.542169
9	(G2_ПОДГУЗНИКИ)	(G2_ТОВАРЫ ДЛЯ КОРМЛЕНИЯ)	0.185185	0.833842	0.284810

Segments Where MBA Does NOT Work

Not all customer segments yield meaningful market basket analysis results. For example, Cluster 0, characterized by one-off purchases, showed no stable A → B rules or frequent category pairs above the support threshold.

> For customers making one-off purchases, popularity-based or price-driven recommendations are more effective than basket rules.

Final Conclusions & Key Takeaways Takeaways

Meaningful associations emerge only after customer segmentation, revealing different real-life shopping scenarios. Using several methods increases confidence in discovered patterns.

Cross-Sell Recommendations

Bundled Offers

Segment-Specific Promotions

Market Basket Analysis is most effective when combined with behavioral segmentation and segmentation and focused on a small number of strong, interpretable rules.